Consensus Pseudo-Labels for Solar Filament Instance Segmentation

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Abstract

Solar filaments are thin, low-contrast dark structures in full-disk solar images; the challenge asks for them as *instances*, not as a foreground mask. The organiser's metric is a row-wise panoptic-quality (PQ) score: every annotator–image row of the ground truth is scored against the same predicted instances, so a prediction must agree with the annotators simultaneously, rather than with one of them. We first attacked the task with a hand-crafted consensus detector, then replaced it by learning. Our pipeline turns the public multi-annotator polygons into per-pixel consensus pseudo-labels (a support count plus a per-class vote), trains a 7.76M-parameter U-Net on 1024×1024 maps, extracts instances by thresholding, a 7×7 morphological closing, 4-connectivity labelling and an area floor with a per-image relaxation fallback, and averages four seeds with horizontal-flip test-time augmentation. On a fixed 88-image validation split (220 annotator–image rows, 1,514 ground-truth masks, 21 annotators) the local replication of the official score rises from 0.2201 to 0.3449, and the public-leaderboard score from 0.18 to 0.29. We also report negative results (class splitting, geometric watershed, a wider network) and the optimism (≈ 0.015 PQ) of using one shared validation split for monitoring, thresholding and seed selection.

CCS Concepts

• **Computing methodologies** → **Image processing**; • **Applied computing** → *Astronomy*.

Keywords

instance segmentation, solar filaments, consensus pseudo-labels, U-Net, panoptic quality, sunspot and filament analysis

1 Introduction

Filaments are magnetic features that appear as dark, elongated structures on the solar disk. Segmenting them is hard for two independent reasons: they are weakly contrasted against the photosphere, and different human annotators trace the same filament differently. The Solar Filament Segmentation Challenge of the IEEE BigData Cup 2026 provides 1,154 annotator–image annotation entries over 707 distinct full-disk images (8,199 single-polygon instances, four category labels: Left, Right, Unidentifiable and Ambiguous), while the 180 test images ship without public annotations. Annotator disagreement is therefore part of the task definition, not noise around it.

The metric makes this explicit. The organiser scores each *annotator–image* row separately and pools the rows: a prediction is a true positive if it overlaps a ground-truth instance of that row with IoU strictly above 0.5, matched pairs are counted (not unique matches), and the score is $\sum \mathrm{IoU(TP)}/(|TP| + 0.5|FP| + 0.5|FN|)$ [1]. A prediction that satisfies only one annotator is a true positive in that row and a false positive in every other row, so the metric rewards shapes that several annotators agree on — the design principle of our second system.

Our first system was a hand-crafted detector that fused several pseudo-annotator scales into an instance-anchored consensus and thresholded a local-contrast image; it reached a local PQ of 0.2201 and a public score of 0.18. Its ceiling was obvious: every improvement came from re-tuning hand-made thresholds. All comparisons below are between our own variants, measured with one protocol on one split; we make no claim about methods we did not run.

We contribute: (i) a consensus pseudo-label construction that is derived from the public polygons only, together with the measurement that motivates it (Section 2.1); (ii) a U-Net recipe and an instance-extraction pipeline with a relaxation fallback (Section 2); (iii) an evaluation protocol that keeps the validation images out of training and quantifies the selection optimism that this shared split introduces (Section 3); and (iv) an ablation table that reports the variants that failed (Section 3.3).

2 Methodology

2.1 Consensus pseudo-labels

The public training polygons are the only supervision available. We rasterised every annotator's polygons onto the native 2048×2048 canvas with the official COCO routine (polygon-to-mask followed by mask merging) and built three per-pixel maps per image: *support* (how many annotators' instances cover the pixel, $0..M$ where M is the number of annotators of that image), *votes* (four channels, one per category, counting the annotators covering the pixel with that category) and *class* (the arg-max of the votes, ties broken in the fixed order Left > Right > Unidentifiable > Ambiguous). The maps are built at the native 2048×2048 resolution and, for the 1024×1024 models, area-averaged in $k \times k$ blocks with $k = 2048/1024 = 2$; the soft target is that average over its per-image maximum, the class map its arg-max over the four channels. The image is resized $2048^2 \rightarrow 1024^2$ bilinearly, so one example is a whole disk, not a sub-crop. The label files also carry a 512×512 view (4×4 block mean of support, block sum of votes) that the earlier 512-pixel model consumed and the 1024-pixel training code never reads. The category statistics of this

release are Left 2,535, Right 2,590, Unidentifiable 3,074 and Ambiguous 0 instances; the Ambiguous channel is kept but is empty, and 3,459,165 pixels are covered by two or more annotators.

Why a consensus instead of a single annotator? The annotations genuinely disagree: over 3,152 greedily matched instance pairs the median IoU is 0.5762 and only 68% of the pairs exceed IoU 0.5. We therefore measured detectors that copy one annotator against detectors that vote, grouping the rows by the number of annotators per image. On two-annotator images, copying an annotator gives PQ 0.6802 and 0.6794, whereas the support ≥ 2 vote gives 0.7090. On three-annotator images the copy baseline averages 0.5556 and the same vote reaches 0.6248. Leave-one-out over the 151 three-annotator images confirms the direction (0.3462 for the union rule against 0.3334 for copying one annotator). The consensus is therefore not a smoothing heuristic: scored against all annotators at once it is closer to them collectively than any single annotation, which is what the row-wise metric rewards.

2.2 Network and training

We train a five-level U-Net with 32 base channels (7,759,653 parameters) on the 1024×1024 consensus maps, with random flips and intensity jitter. The optimiser is Adam with a learning rate of 10^{-3} on a cosine schedule, mixed precision, batch size 4 and 50 epochs; the loss is a soft binary cross-entropy on the support map plus $0.5\times$ cross-entropy on the class map. A base-32 run takes 28.95–30.16 minutes on a single Tesla V100S (155 optimiser steps per epoch). We ran 15 such runs over 13 distinct seeds (the two diverged seeds 7 and 15 were re-run); 6 of them diverged with a non-finite loss and are reported as such rather than silently dropped.

2.3 Instance extraction

At inference the 1024×1024 probability map is bilinearly upsampled to the native 2048×2048 canvas, on which every step below acts: the map is thresholded at 0.70, closed with a 7×7 morphological kernel, labelled with 4-connectivity, and components smaller than 600 pixels are discarded. Because an empty prediction for an image is catastrophic under a row-wise metric, an image that would end up empty is reprocessed along a relaxation chain (threshold 0.55 with area floor 200, then threshold 0.45, then the largest component alone), so that no test image is left without an instance. On the submitted ensemble, 5 of the 180 test images needed that fallback and the produced file passed the submission contract check with zero issues.

2.4 Ensembling

Four independently trained seeds (three with the same recipe but different seeds, plus the seed used for the single-model submission) are averaged in probability space, each with horizontal-flip test-time augmentation, and post-processed exactly as above.

Table 1: Validation results on the fixed 88-image split, using the local replication of the official metric (PQ pooled over annotator–image rows, Dice pooled over pixels; TP/FP/FN in the official convention). TTA = horizontal-flip test-time augmentation.

System	PQ	Dice	TP	FP	FN
Hand-crafted detector	0.2201	0.4861	455	698	1,059
U-Net, 512 px	0.2696	0.6216	603	636	911
U-Net, 1024 px + TTA	0.3152	0.6409	704	637	810
4-seed probability average	0.3370	0.6512	758	654	756
+ 7×7 closing (submitted)	**0.3449**	**0.6547**	779	632	735

3 Evaluation

3.1 Protocol

All local numbers use a local replication of the organiser's self-evaluation notebook [1]: IoU>0.5 strictly, matched pairs counted, numerators accumulating the IoU of the matched pairs, pooled over all annotator–image rows; we also report the pooled pixel Dice.

The split is fixed for every experiment. Of the 707 training images, 296 carry at least two annotators; 88 of them form the validation set (stratified by era and by annotator count, shuffled with a fixed seed), and the remaining 619 images are the training set. The two sets are disjoint (0 shared file names). The validation set contains 220 annotator–image rows, 1,514 ground-truth masks and 21 annotators. The models are trained on the 619 images only; the 88 validation images are never trained on, but they are used for checkpoint selection and for every hyper-parameter scan below, which we quantify in Section 3.6.

3.2 Main results

Table 1 lists the progression of our own systems under one protocol. The hand-crafted consensus detector reaches PQ 0.2201 and Dice 0.4861. Learning from consensus pseudo-labels gives 0.2696 / 0.6216 with a 512-pixel model; a 1024-pixel model with flip TTA gives 0.3152 / 0.6409. Averaging four seeds adds +0.0218 PQ (0.3370 / 0.6512) and the morphological closing a further +0.0079 (0.3449 / 0.6547), with 554 predicted instances over the 88 images. The 512-pixel row is an earlier and much smaller model (0.48M parameters, 90 epochs, 14.5 minutes on an RTX 3070 laptop GPU) whose training log records 473 training and 85 validation images: at that time 558 of the 707 training images had been downloaded, so it saw 473 images against the 619 used by the 1024-pixel models; the step to 1024 pixels therefore changes resolution, capacity, schedule and data size together and is not an ablation of resolution alone.

3.3 Ablations, including the negative ones

Table 2 summarises the variants we tested. Three of them are negative results that we report because they shaped the final design. Splitting an instance by its Left/Right class label costs −0.0224 PQ (0.3142 to 0.2917): the split produces a narrower fragment that misses the annotator boundary, and the metric punishes both the new false positive and the new

Table 2: Ablations on the same split, with the same probability maps and evaluator; Δ (in brackets) is the change in PQ with respect to the row it is compared against; TTA = horizontal-flip test-time augmentation, k = closing kernel.

Variant	PQ	TP/FP/FN
TTA off (1 seed)	0.2922 [−0.0230]	689/772/825
TTA on (1 seed)	0.3152	704/637/810
Closing $k = 3/5/7/9/13$	0.3386–0.3449	—
Class split (Left/Right)	0.2917 [−0.0224]	649/669/865
Watershed (best of 4)	0.1123 [−0.2019]	286/1,427/1,228
Wider net (48 base ch.)	0.2096 [−0.0870]	494/942/1,020
Best single seed	0.3268	—
4-seed avg., no closing	0.3370 [+0.0102]	758/654/756
4-seed avg. + closing	0.3449 [+0.0079]	779/632/735

false negative. Replacing connected components by a geometric watershed is much worse (−0.2019, 0.3142 to 0.1123): the watershed over-segments the foreground into hundreds of fragments and the false-positive count grows from 635 to 1,427. Widening the network from 32 to 48 base channels also hurts: the base-48 run reaches 0.2096 against 0.2966 for the base-32 run of the same family. The two runs differ in seed, so the comparison also carries the seed spread quantified below, but this is the only wider configuration we trained, and it ends 0.087 PQ behind while costing 2.25× the parameters and 48.7 instead of 29.0 minutes per seed, so we abandoned that line.

Two further observations qualify the remaining gains. First, the closing kernel is not a knife-edge: kernels 7, 9 and 13 give 0.3449, 0.3443 and 0.3441, a spread of 0.0008, so the gain comes from closing the gaps rather than from the particular kernel size. Second, the seed spread is much larger than the ensembling gain: over the four seeds for which we have complete sweeps, the best validation PQ ranges from 0.2070 to 0.2905 without flip TTA and from 0.2231 to 0.3142 with it, i.e. an envelope of 0.2070–0.3142, while the four-seed average gains +0.0218 over the single-seed system it replaces. Seed selection therefore carries a risk that ensembling only partly offsets.

3.4 Public leaderboard

Eight files have been submitted. Their public scores rise together with our own local scores: 0.15 (first hand-crafted file, 126 of 180 test images covered), 0.16, 0.17 and 0.18 (the hand-crafted consensus detector, 174 of 180 images covered), then 0.21 for the 512-pixel U-Net (local validation PQ 0.2696), 0.25 for a two-seed average (0.2993), 0.27 for the single 1024-pixel seed (0.3152) and 0.29 for the submitted four-seed ensemble with closing (0.3449). Dividing the public score by the local validation PQ of the same file gives 0.779, 0.835, 0.857 and 0.841 for the four model submissions, i.e. a band of 0.78 to 0.86, which indicates that the local protocol ranks our own variants in the same order as the public board does. We stress two caveats. The public score is a coarse signal, not an estimate of our test performance: the API metadata announces a panoptic-quality algorithm while the challenge page describes a mean-Dice score, and the leaderboard exposes a single score column, so the two can never be reconciled from the outside. And the local numbers behind those ratios are themselves optimistic, for the reason given next.

3.5 Reproducibility

Every number quoted here is read from a retained evaluation artifact rather than transcribed: the per-configuration evaluation files (`eval_*_val88.json`), the seed table (`seed_pq_table.json`, 15 training runs), the post-processing sweep (`postproc_sweep_val88.csv/.json`, 60 variants), the ensemble sweeps (`ens_specs_val88.csv`, `custom_specs_val88.csv`, 64 and 28 configurations), the submission reports (`submission_*_report.json`), the captured public-leaderboard listing (`kaggle_submissions_20261101.txt`, 8 submissions) and the training log of the 512-pixel model (`train_v1.log`). The file `report_evidence.json` maps the evaluation numbers quoted here to their source artifact.

3.6 Limitations

Four limitations qualify everything above. *(i) Selection optimism.* The 88 validation images play three roles: they monitor training and pick the checkpoint, they select the threshold, the closing kernel and the area floor over a 60-variant post-processing scan, and they select the seed. Every validation number in this paper is therefore a post-selection maximum. We quantified the effect by selecting the best variant on one half of the split and evaluating it on the other half: the held-out mean is 0.3019 against 0.3172 for the best variant chosen on the full split, i.e. an optimism of about 0.015 PQ, and the estimate ranges from +0.002 to +0.015 across four different splits of the validation set. The same applies to the headline 0.3449, which was also chosen on this split. *(ii) Metric ambiguity.* The official description of the leaderboard score is inconsistent between the API metadata and the challenge page, so we use the public score only for ordering. *(iii) Seed variance.* The spread between the four seeds with complete sweeps (0.2070–0.3142) is larger than most of the individual improvements we report, and 6 of 15 training runs diverged with a non-finite loss, which shows that the optimisation itself is fragile. *(iv) Data scale.* The supervision is limited to 707 public training images; the 512-pixel model was trained on 473 of them, since only 558 were on disk at the time and 85 of those were held out for validation, whereas the 1024-pixel models used all 619 images of the training split, so the effect of the larger label set is not isolated.

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originate from the NSO/GONG network. We also thank the annotators whose multiple polygon drawings per image make the consensus analysis in Section 2.1 possible.

References

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